IT-Security-Project

Project Overview

This project focuses on performing a basic security audit on a Windows system to evaluate its security posture and identify potential vulnerabilities.

The audit focuses on the following areas:

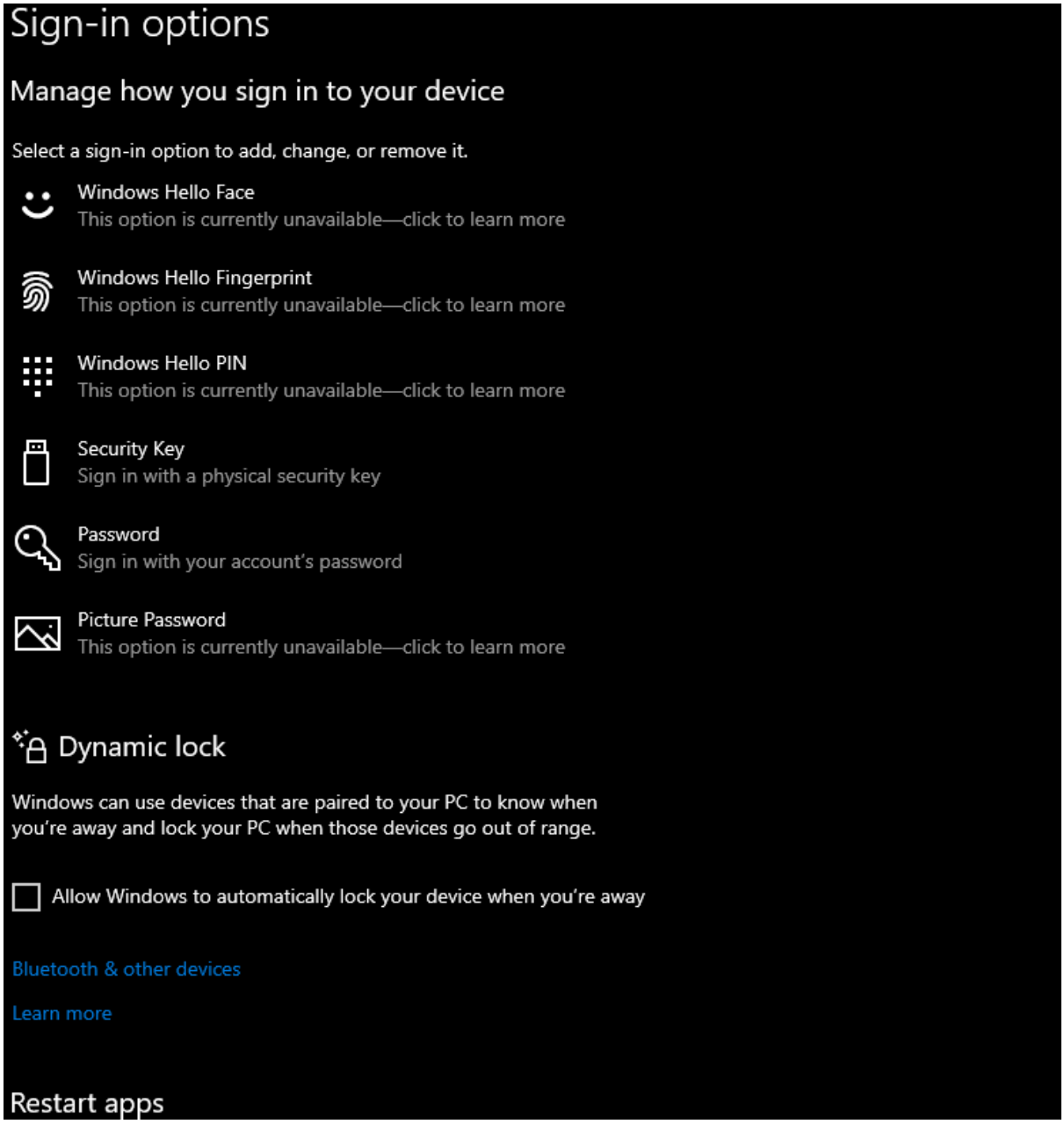
- Password strength and authentication policies
- Firewall configuration
- System and software updates

These checks help ensure the system is protected from unauthorized access, malware, and other cybersecurity threats.

Practical Task: Security Audit

Step 1: Check Password Strength

Strong passwords help prevent unauthorized access to systems.

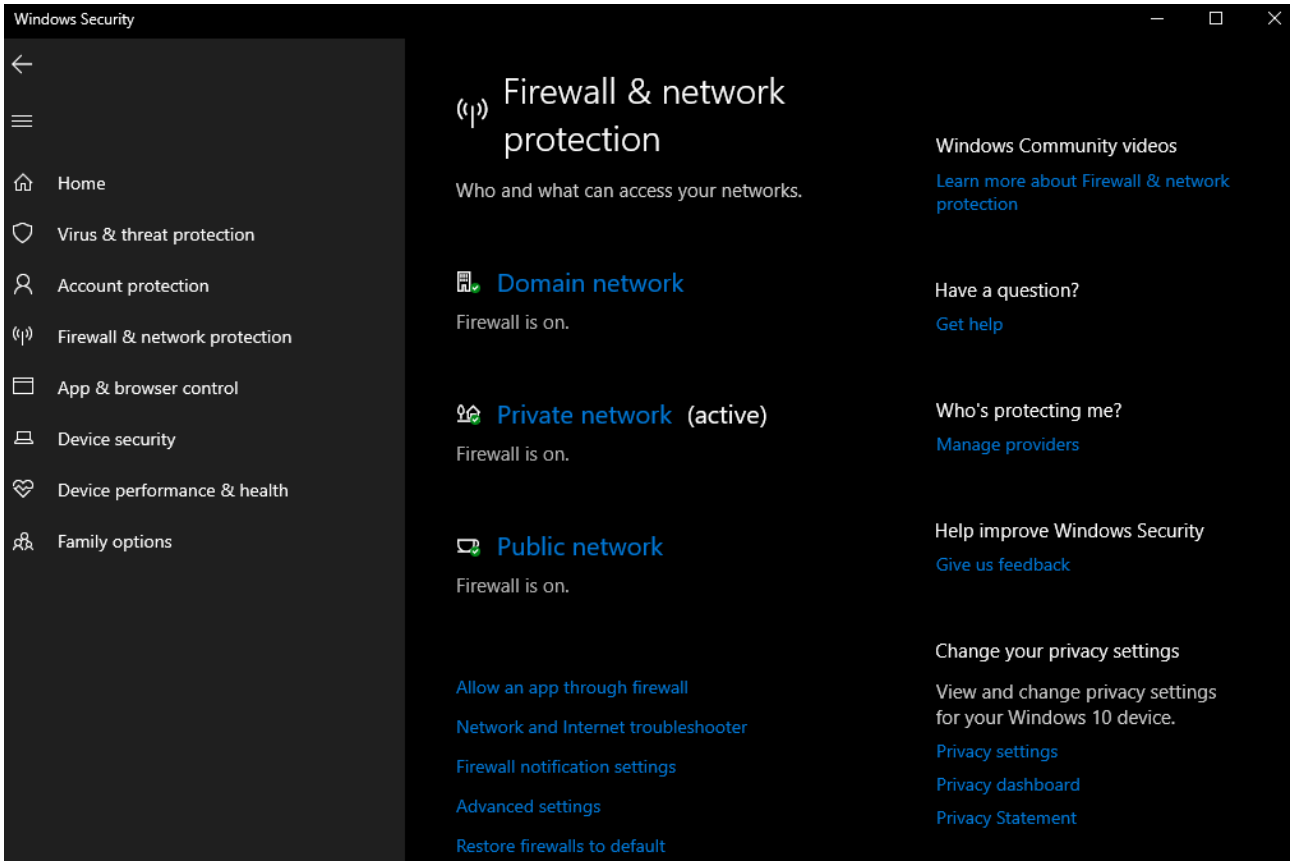


Password Checklist

- Minimum password length ≥ 8 characters
- Use uppercase letters
- Use lowercase letters
- Include numbers
- Include special characters
- Avoid common or easily guessable passwords

Step 2: Check Firewall Status

A firewall protects the system by blocking unauthorized network access.



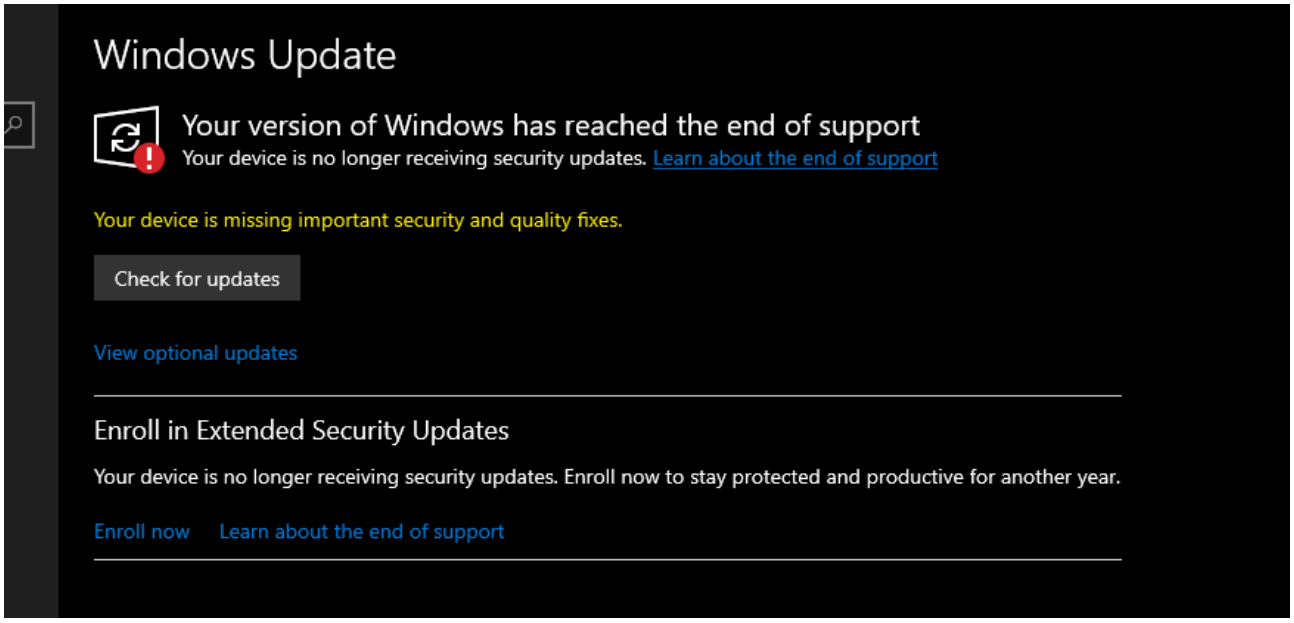
Screenshot:

Firewall Checklist

- Firewall is enabled
- Firewall is active on all network profiles
- Only necessary services and ports are allowed

Step 3: Check System and Software Updates

Keeping the operating system updated helps fix security vulnerabilities.



Update Checklist

- All critical security updates installed
- System running the latest security patches
- Automatic updates enabled

Document Findings

After performing the security checks on the system, the results were documented to track the security status and provide recommendations for improving system protection.

Check	Status	Suggestions
Password Strength	✅ Strong / ⚠ Weak	Update weak passwords to strong, complex ones
Firewall Status	✅ Enabled / ⚠ Disabled	Keep firewall ON for all profiles
System Updates	✅ End of Support/ ⚠ Devices no longer receives security updates	Upgrade to a supported Windows version or enroll in ESU

Visual Security Checklist Report

Password Strength

Status:

Weak passwords found / ✅ Strong passwords implemented

Suggestions for Improvement

- Use passwords with 12 or more characters
- Include uppercase letters, lowercase letters, numbers, and symbols
- Avoid dictionary words or repeated sequences

- Enable account lockout after multiple failed login attempts
- Consider enabling Multi-Factor Authentication (MFA)

Firewall Status

Status:
Firewall enabled on all profiles

Suggestions for Improvement

- Keep the firewall enabled for Domain, Private, and Public profiles
- Regularly review firewall logs
- Ensure firewall rules update automatically when supported

System Updates

Status:
End of Support/ ✅ Upgrade to a supported Windows version or enroll in ESU

End of Windows Support

- The installed version of Windows has reached End of Support (EoS).
- The system no longer receives security updates.
- This increases the risk of cyber-attacks, malware, and ransomware.

Missing Security and Quality Updates

- The device is missing important security updates.
- Unpatched vulnerabilities may be exploited by attackers.
- System performance and stability may be affected.

Extended Security Updates (ESU)

- Windows offers an option to Enroll in Extended Security Updates (ESU).
- ESU provides temporary security updates after support ends.
- This is only a short-term solution (usually up to one year).

Conclusion

Regular system security audits help identify vulnerabilities and strengthen system protection. Implementing strong passwords, maintaining firewall protection, and ensuring systems are updated are key practices for maintaining a secure computing environment. This report was prepared by Nthabiseng Tsunke as part of the IT Security course assignment.